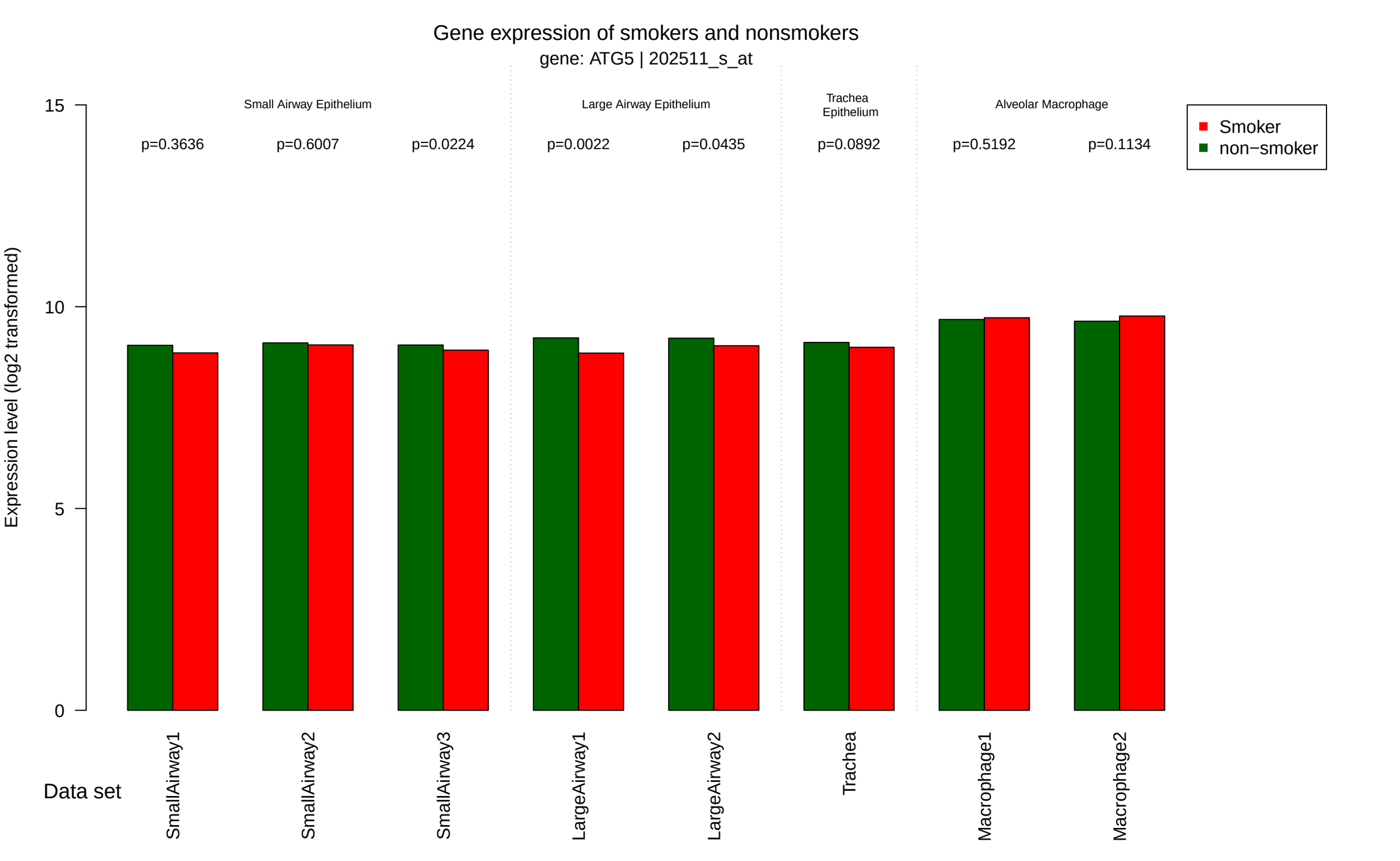
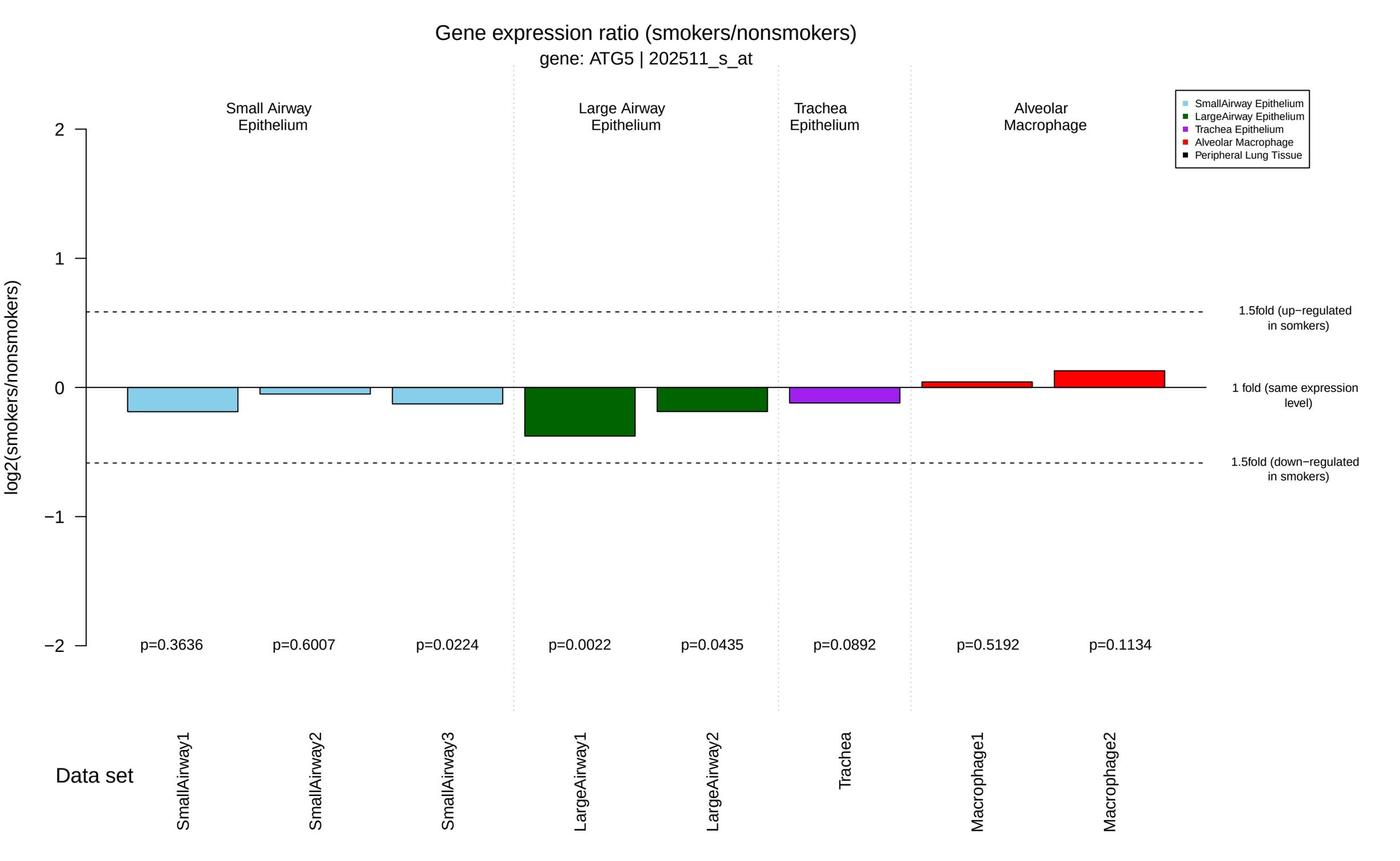
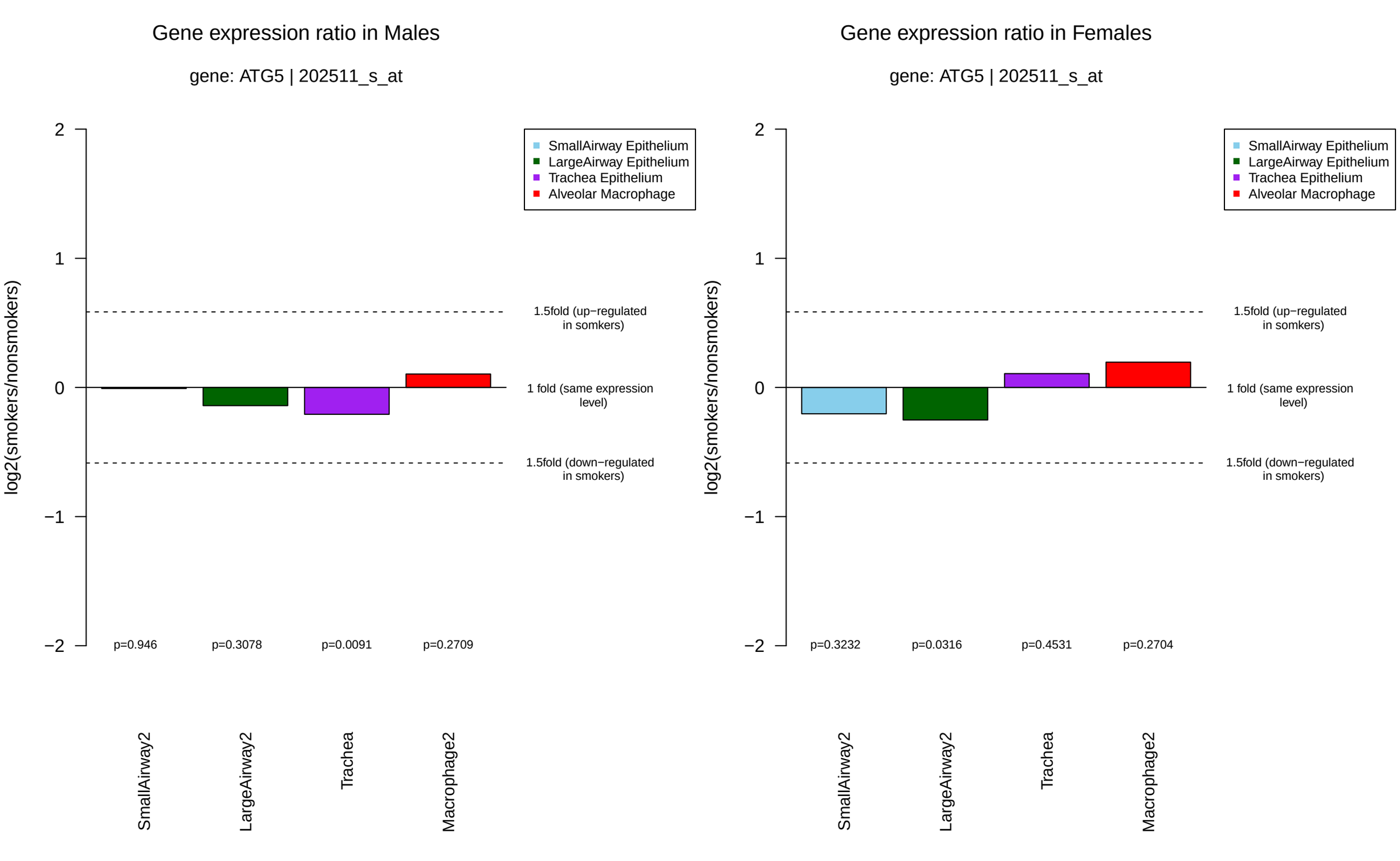
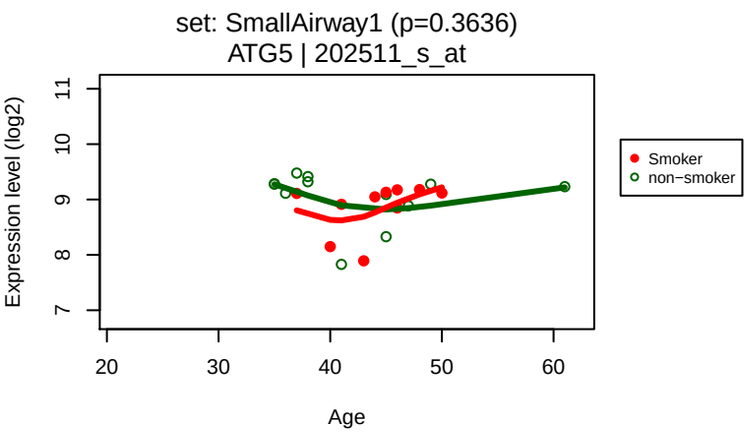
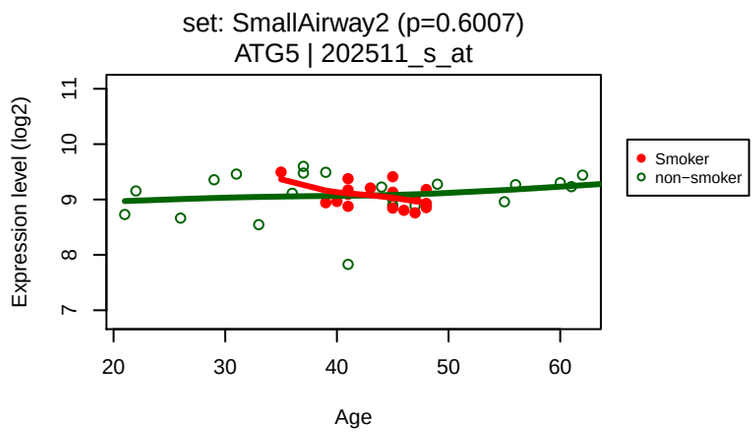
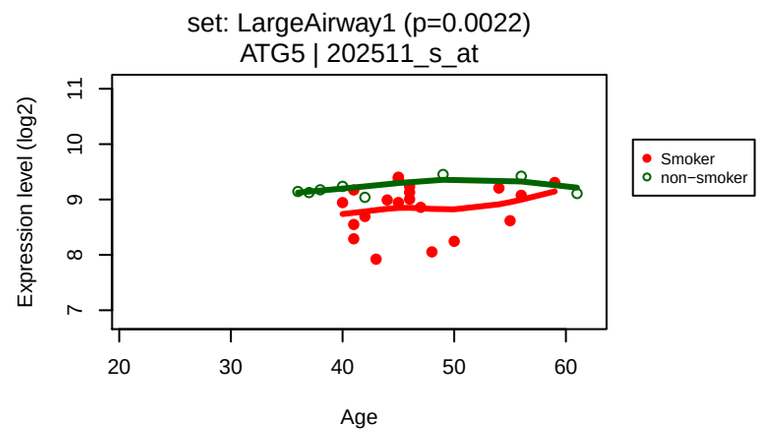

Gene expression of smokers and nonsmokers
gene: ATG5 | 202511_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3636
p=0.6007
p=0.0224
p=0.0022
p=0.0435
p=0.0892
p=0.5192
p=0.1134
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

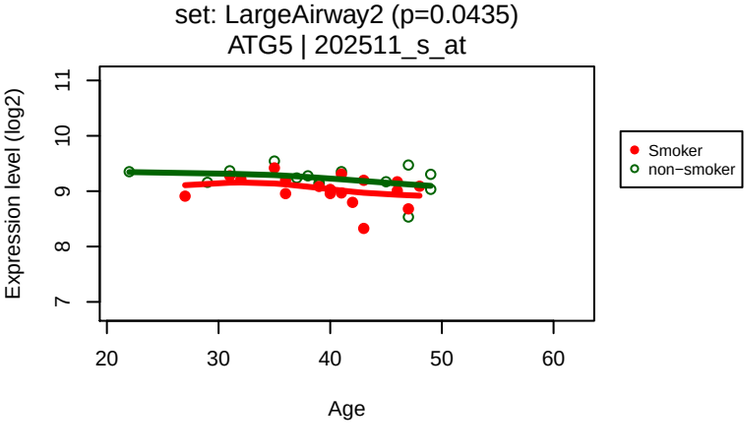

Gene expression ratio (smokers/nonsmokers)
gene: ATG5 | 202511_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3636
p=0.6007
p=0.0224
p=0.0022
p=0.0435
p=0.0892
p=0.5192
p=0.1134
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

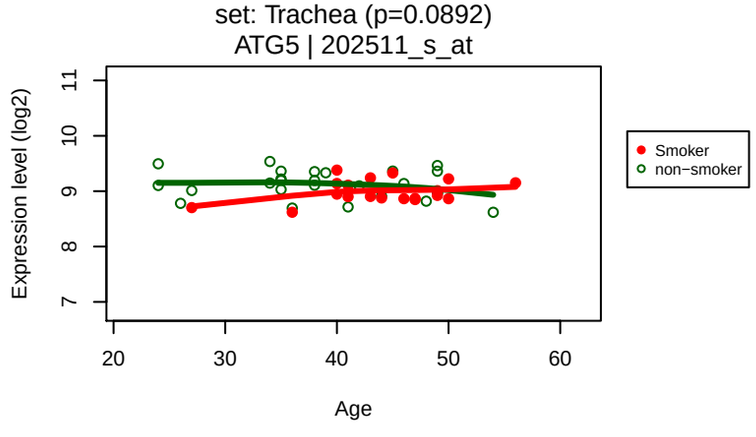

Gene expression ratio in Males
gene: ATG5 | 202511_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.946
p=0.3078
p=0.0091
p=0.2709
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATG5 | 202511_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3232
p=0.0316
p=0.4531
p=0.2704
SmallAirway2
LargeAirway2
Trachea
Macrophage2

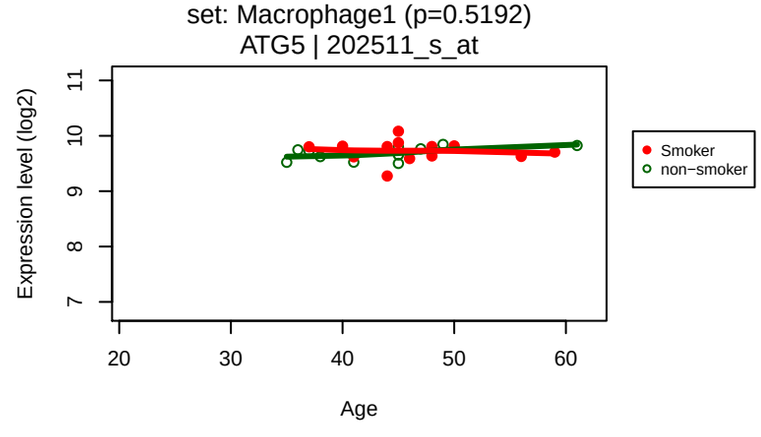

set: SmallAirway1 (p=0.3636)
ATG5 | 202511_s_at
Expression level (log2)
Age
Smoker
non−smoker

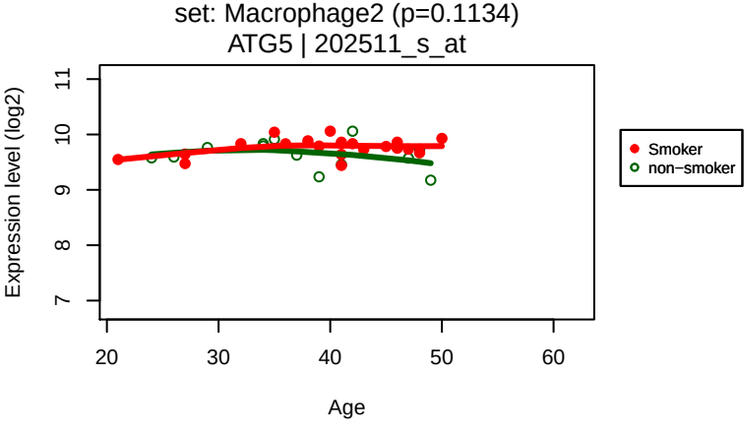

set: SmallAirway2 (p=0.6007)
ATG5 | 202511_s_at
Expression level (log2)
Age
Smoker
non−smoker

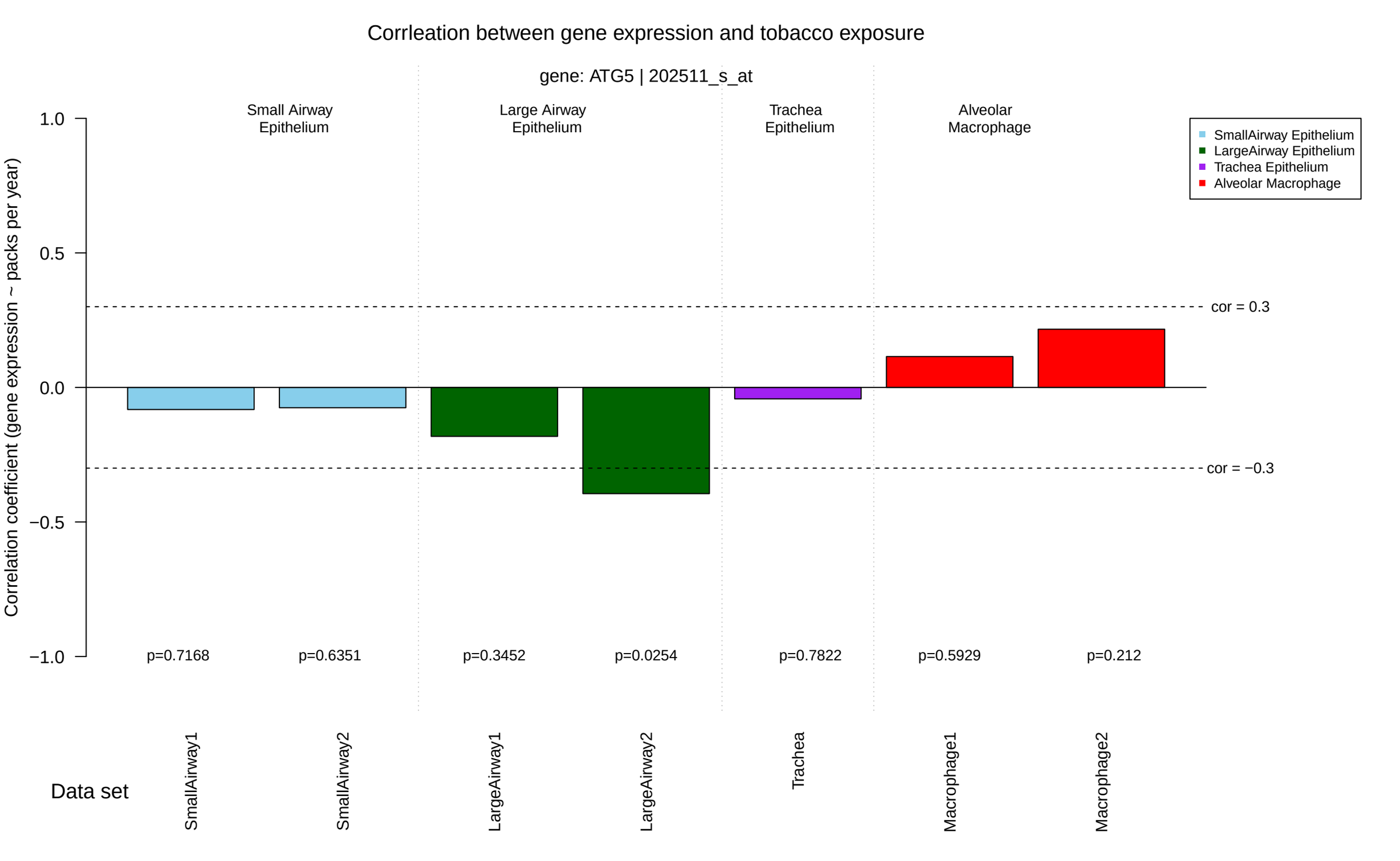

set: LargeAirway1 (p=0.0022)
ATG5 | 202511_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0435)
ATG5 | 202511_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0892)
ATG5 | 202511_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5192)
ATG5 | 202511_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1134)
ATG5 | 202511_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATG5 | 202511_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7168
p=0.6351
p=0.3452
p=0.0254
p=0.7822
p=0.5929
p=0.212
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2